Linking Moles, Mass & Mr

Exam code: 0620 & 0971

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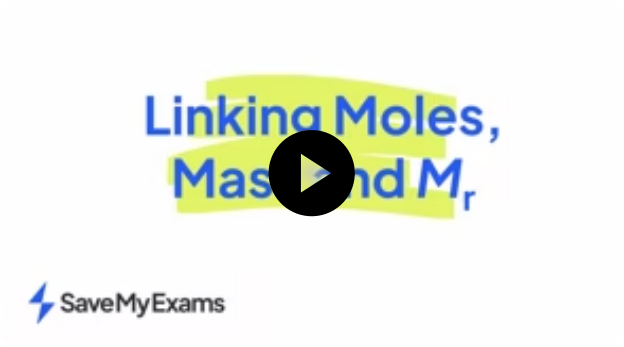

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Linking moles, mass & Mr

Extended tier only

- One mole of any element is equal to the relative atomic mass of that element in **grams**
- If you had 1 mole of carbon atoms in your hand, that number of carbon atoms would have a mass of 12 g (because the A_r of carbon is 12)
  - So, one mole of helium atoms has a mass of 4 g (A_r of He is 4)
  - One mole of lithium has a mass of 7 g (A_r of Li is 7) and so on
- One mole of any compound is the **relative molecular mass** or relative formula mass in **grams**
- To find the mass of one mole of a compound, add up the relative atomic masses
  - For example, carbon dioxide has an *Mr* of:

$$(1 \times C) + (2 \times O)$$

$$(1 \times 12) + (2 \times 16) = 44$$

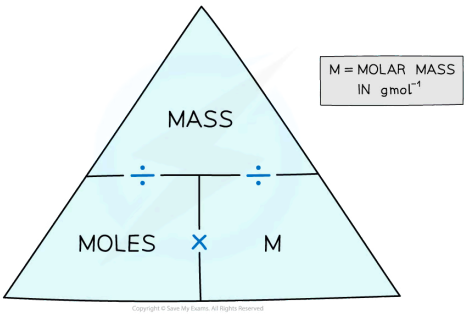
Moles, mass and relative mass

- The number of moles of any chemical can be calculated using:

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

between moles, mass in grams and the molar mass:



Formula triangle for moles, mass and molar mass

- Calculating the number of moles of an element uses the same equation, but with relative atomic mass replacing M

Worked Example

What is the mass of 0.250 moles of zinc?

Answer:

- From the Periodic Table, the relative atomic mass of Zn is 65
  - So, the molar mass is 65 g / mol
- The mass is calculated by **moles x molar mass**:

0.250 mol x 65 g / mol = **16.25 g**

Worked Example

How many moles are in 2.64 g of sucrose, $C_{12}H_{22}O_{11}$ (M_r = 342)?

Answer:

- The molar mass of sucrose is 342 g / mol
- The number of moles is found by **mass ÷ molar mass**:

$$\frac{2.64}{342} = \mathbf{7.72 \times 10^{-3}\ mol}$$

Examiner Tips and Tricks

Always show your workings in calculations as its easier to check for errors and you may pick up credit if you get the final answer wrong.

Calculating moles & masses

- Chemical amounts are measured in moles
- The mole, symbol mol, is the SI unit of **amount of substance**
- One mole of any substance contains the same number of the stated particles
  - This can be atoms, molecules or ions
- One mole contains **6.02×10^{23}** particles
  - This number is known as the **Avogadro constant**
- For example:
  - One mole of sodium (Na) contains 6.02×10^{23} atoms of sodium
  - One mole of hydrogen (H_2) contains 6.02×10^{23} molecules of hydrogen
  - One mole of sodium chloride (NaCl) contains 6.02×10^{23} formula units of sodium chloride

Worked Example

For magnesium chloride, $MgCl_2$, calculate the number of:

1. Molecules in 1 mole
2. Atoms in 1 mole
3. Chloride ions in 1 mole
4. Magnesium ions in 2 moles

Answers:

1. The formula is $MgCl_2$, so 1 mole of $MgCl_2$ is:

$$1 \times 6.02 \times 10^{23} = 6.02 \times 10^{23} \text{ molecules}$$

2. There are 3 atoms in $MgCl_2$, so 1 mole of $MgCl_2$ contains:

$$3 \times 6.02 \times 10^{23} = 1.806 \times 10^{24} \text{ atoms}$$

3. There are 2 chloride ions in $MgCl_2$, so 1 mole of $MgCl_2$ contains:

$$2 \times 6.02 \times 10^{23} = 1.204 \times 10^{24} \text{ chloride ions}$$

4. There is 1 magnesium ion in $MgCl_2$, so 2 mole of $MgCl_2$ contains:

$$2 \times (1 \times 6.02 \times 10^{23}) = 1.204 \times 10^{24} \text{ magnesium ions}$$

Worked Example

In 15.7 g of water ($M_r = 18$):

1. How many molecules are there?
2. How many atoms are there?

- The molar mass of water is 18 g / mol
- The number of moles is found by **mass ÷ molar mass**

15.7 g ÷ 18 g / mol = **0.872 mol**

- There are 6.02×10^{23} molecules of water in 1 mole of water
- So, in 0.872 moles of water, there are:

$6.02 \times 10^{23} \times 0.872 =$ **5.25×10^{23}** molecules

The number of atoms:

- In each molecule of water, there are 3 atoms (2 hydrogen atoms, one oxygen atom)
- So, the number of atoms in 15.7 g is:

$3 \times 5.25 \times 10^{23} =$ **1.58×10^{24}** atoms

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